

Research Proposal

Title

Phenological Monitoring of Almond Trees: Integrating UAV Imagery and AI to Predict Bloom Timing and Yield

Literature Review

Almond trees follow a well-defined phenological cycle that begins with flower bud initiation in late summer or early autumn, followed by bud dormancy during the winter months. As temperatures rise, bud swelling and bud break occur, typically in late winter. This is followed by flowering (bloom), often in early spring, which includes full bloom and petal fall stages. After successful pollination, the trees enter the fruit set stage, transitioning into nut development, shell hardening, and kernel filling throughout the spring and summer. The cycle concludes with hull split and eventual harvest in late summer to early autumn, after which the cycle restarts with preparation for the next season's bud initiation. This flower cycle is tightly regulated by both genetic predisposition and environmental conditions, particularly temperature fluctuations, photoperiod sensitivity, and chill hour accumulation ([R Sánchez-Pérez, et al., 2014](#); [Zhang et al., 2025](#)). Notably, almonds differentiate from other *Prunus* species in their ability to develop flower buds on both perennial spurs and current-year shoots, thereby increasing their reproductive capacity and stabilizing yields ([Prudencio et al., 2020](#)).

The precise timing of bud development, dormancy release, and bloom onset plays a pivotal role in orchard productivity. The flowering season in almond trees typically spans from January to March. Early flowering increases the risk of frost damage to buds and blossoms, while delayed bloom may reduce pollination efficiency due to temporal mismatches with pollinator activity, ultimately leading to yield losses ([Yaacoubi et al., 2019](#)). These phenological stages spanning dormancy, bud swell, anthesis, fruit set, and harvest are profoundly influenced by cultivar-specific chilling and heat requirements as well as environmental variability and orchard management strategies ([García-Gusano et al., 2004](#); [Campoy et al., 2011](#)). Recent long-term observations in Mediterranean climates have demonstrated that warmer winters can delay dormancy release due to insufficient chilling, while elevated spring temperatures may accelerate bloom once chilling is satisfied ([Yaacoubi et al., 2019](#)). Consequently, reliable phenological monitoring through both observational tools and predictive models is essential for adaptive decision-making in both

research and commercial almond production. Accurate detection and monitoring of flowering phenology is critical for yield prediction in almond orchards, as the timing, intensity, and uniformity of bloom directly influence pollination success, fruit set, and eventual kernel development. Early detection of asynchronous or delayed flowering, often driven by climatic variability, can signal potential yield reductions. Therefore, high-resolution monitoring of bloom stages using drone-based imaging and AI-powered phenological analysis provides a powerful tool for optimizing orchard management, predicting yield variability, and mitigating climate-related risks.

Quantifying flower density in almond orchards is essential for accurate yield estimation and effective management decisions. Since almond yield is influenced more by the total number of flowers than by the proportion that successfully set fruit, early assessment of bloom intensity provides a more reliable predictor of final productivity ([Tombesi et al., 2017](#)). This makes flower abundance a key variable for anticipating yield potential and guiding in-season interventions. In parallel, advances in sensor technologies now enable precise and scalable flower quantification. Using LiDAR and RGB imaging, [Underwood et al. \(2016\)](#) demonstrated that flower density measurements could be automated and spatially mapped at the tree level, supporting large-scale monitoring and improving the efficiency of resource allocation. Together, these findings underscore the role of flower quantification as a cornerstone of data-driven almond orchard management.

Conventional flower counting methods such as manual tagging, branch sampling, and visual estimation are labor-intensive, subjective, and unsuitable for large-scale orchard monitoring. Recent advancements in remote sensing and computer vision have enabled scalable and objective alternatives. In particular, the use of aerial RGB imagery combined with deep learning techniques allows for high-throughput, spatially detailed assessments of bloom intensity. [Chakraborty et al. \(2023\)](#) demonstrated that integrating aerial bloom mapping with convolutional neural networks enables accurate, spatially explicit yield predictions early in the growing season. This approach not only reduces the time and labor associated with traditional sampling but also improves the reproducibility and spatial resolution of phenological monitoring, reinforcing its value for precision agriculture and data-driven orchard management.

Background and Motivation

Flowering dynamics in almond orchards are shaped by genetic, climatic, and agronomic factors, with chill accumulation playing a pivotal role in bud break and bloom synchronization ([Zhang et al., 2025](#)). Conventional monitoring approaches, such as visual scoring a method where trained observers directly estimate the quantity or developmental stage of blossoms on trees using visual inspection and predefined rating scales and manual flower counts are limited in scale and precision. In the face of climate uncertainty and increasing pressure to optimize pollination and inputs, the almond industry requires objective, scalable, and cost-effective methods for flower quantification.

Despite its importance, a key knowledge gap remains in the accurate, scalable, and timely detection of flowering in almond orchards using unmanned aerial vehicles (UAVs). Traditional field-based floral surveys are labor-intensive, subjective, and often infeasible across large or heterogeneous orchards. While UAVs offer high spatial and temporal resolution, automated flower detection remains challenging due to variable lighting, overlapping canopies, spectral similarity between flowers and background elements, and occlusion by leaves or branches. Moreover, current models often lack generalizability across varieties, growth stages, and environmental conditions.

Recent developments in UAVs and deep learning offer a viable solution. By leveraging high-resolution RGB imagery and convolutional neural networks, researchers can identify flowering zones, assess phenological variability, and integrate these data with environmental indicators to inform yield prediction and resource allocation ([Dias et al., 2018](#)). Addressing this gap requires the development of robust, adaptable computer vision and AI frameworks capable of detecting and quantifying flowers under real-world orchard conditions, which is crucial for improving yield prediction and decision-support systems in precision agriculture. Despite recent progress in image-based flower quantification, key challenges remain. Many existing models are developed using datasets limited to specific seasons, locations, or orchard conditions, which limits their ability to generalize across varying climates and almond cultivars. Moreover, most approaches do not incorporate essential environmental variables such as chill hour accumulation that play a central role in determining flowering behavior and yield potential ([Zhang et al., 2025](#); [Prudencio et al., 2014](#)). Addressing these gaps requires the development of integrated methodologies that combine high-resolution bloom mapping with detailed phenological and climatic data. Such models would offer greater predictive accuracy and broader applicability, ultimately improving both scientific insight and on-the-ground decision-making in almond orchard management.

Research Objectives

This study aims to bridge traditional phenological observations of almond flowering with UAV- and machine learning–driven monitoring, and to test its relationship to yield through the integration of high-resolution aerial imagery, computer vision algorithms, and ground-based yield measurements, through the following goals:

1. **Develop and validate** a deep learning pipeline for almond tree segmentation using U-Net and comparative models on UAV-acquired RGB imagery.
2. **Estimate flowering levels** using pixel-based blooming index (EBI) thresholding within segmented canopies, accounting for variable lighting conditions.
3. **Classify orchard rows** into pollinator and pollinated groups using flowering estimates and assess synchronization.
4. **Integrate historical chill hour data** to analyze spatial relationships between thermal accumulation and flowering patterns.
5. **Benchmark drone-based flowering estimation** against ground truth yield observations.

Research Questions

- How accurately can segmentation models applied to UAV-derived imagery quantify almond flowering intensity at the tree or canopy level?
- What thresholding strategy best captures white-petal regions blooming index of flowering in RGB images?
- How can we incorporate chill hour data with segmentation-based models into a unified framework for almond flowering estimation?
- How does chill hour data improve the predictions of flowering intensity and distribution?
- How does flowering synchronization between almond cultivars affect overall yield?

Methodology

A multi-season, multi-site UAV dataset of high-resolution RGB images will be collected across almond orchards at key flowering stages. Tree-level annotations (manual flower counts and canopy boundaries) will be acquired to enable supervised learning and validation. a U-Net convolutional neural network will be trained for tree crown segmentation ([Ronneberger et al., 2015](#)), with data augmentation techniques such as affine transformations, histogram equalization, and brightness adjustments to ensure model robustness under varying field conditions. Performance will be evaluated using Intersection over Union (IoU) and Dice coefficient scores. For flower segmentation, Mask R-CNN ([He et al., 2017](#)) and additional architectures will be benchmarked.

Flowering estimation will be based on pixel-level color classification, calculating the proportion of white or light-pink petal pixels within each segmented tree canopy using HSV and calibrated RGB thresholding techniques. To enhance sensitivity to bloom intensity, blooming indices such as the Excess Blooming Index (EBI) will also be applied, capturing subtle variations in floral density and distribution ([Chen et al., 2019](#)). These indices and thresholds will be fine-tuned using annotated imagery and validated against flower counts.

To explore environmental drivers, chill hour accumulation data (using the Utah or Dynamic Model; [Anderson, Richardson, & Kesner, 1986](#); [Erez, Couvillon, & Hendershott, 1979](#)) from local weather stations will be spatially linked to tree-level flowering intensity. Regression and correlation analyses will assess the predictive power of chill hours in flowering timing and density. To evaluate flowering synchronization, results will be aggregated by cultivar (pollinator vs. pollinated rows), and cross-correlation analyses will be applied to assess the overlap in bloom periods. This will be linked to yield data to assess how cultivar-level flowering synchrony affects productivity ([Rodríguez-Pérez & Traveset, 2016](#)). Finally, the UAV-based flowering estimates will be benchmarked against traditional tree based yield outcomes to evaluate system performance and potential for operational deployment in precision pollination and yield forecasting.

Expected Outcomes

- Robust deep learning pipeline for almond trees segmenting and flowering estimation based on drone imagery, effective across diverse seasons and locations.
- Higher accuracy and scalability in flower estimation compared to manual and other automated methods, validated with field data and standard metrics.

- Integrated phenological analysis linking flowering patterns to chill hour data, improving understanding of factors affecting bloom time and yield.
- Guidance for future research, including adapting these methods to other crops and combining with satellite remote sensing for broader monitoring.

Timeline

Task	May-Jun 2025	Jul-Sep 2025	Oct-Dec 2025	Jan-Apr 2026	May-Jul 2026
Data preparation & annotation	X				
Model development (segmentation)		X			
Flowering estimation pipeline		X	X		
Row-level classification		X	X		
Integration with chill hour data			X	X	
Validation & comparison				X	
Linking flowering patterns to chill hour data to yield				X	
Writing & thesis submission				X	X

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